

Software Engineering

Mini Project

Project Proposal

Electronics Store Inventory Management System

Group Number: 16

Prepared by

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1. Project Title and Group Number

Project Title: Electronics Store Inventory Management System

Group Number: 16

2. Team Profile

S.No	Name	SRN
1	Shreya C Rao	PES2UG24CS481
2	Shreyas Gowda K A	PES2UG24CS490
3	R Laksheeth	PES2UG25CS819

3. Proposed Project Description

The Electronics Store Inventory Management System (ESIMS) is a web-based application designed to help an electronics store maintain, monitor and manage its inventory efficiently. It provides a centralized database for electronic components/products and allows authorized personnel to add, update, delete, search and view inventory records, maintain the current quantity available for each product, identify availability status, and generate reports. A dashboard provides a concise view of overall inventory health.

The system addresses problems associated with manual or spreadsheet-based inventory management, including duplicate records, inaccurate stock counts, difficulty locating components, delayed identification of low-stock items and lack of a consolidated inventory view. The system is proposed as a new, self-contained academic software project.

Version 1.0 scope: The system maintains the current quantity available for each product. It does not maintain detailed Stock In/Stock Out transaction history or transaction records.

Typical Users

- **Administrator:** Manages users and has full access to inventory operations and reports.
- **Store Manager:** Performs inventory CRUD operations, updates current quantities and monitors inventory.
- **Store Employee:** Views/searches inventory and performs permitted routine inventory operations.
- **Management:** Uses dashboard summaries and availability reports for decisions.

Functional Features

- Secure login and role-based access control.
- View and search the product inventory by product ID, name, or category.
- Add a new product to inventory with details such as product ID, name, category, price, quantity, reorder level and location.
- Delete an existing product record from the inventory.
- Update the current quantity or details of an existing product.
- Prevent duplicate product IDs, with input validated at the point of entry.
- Automatically classify each product as In Stock, Low Stock, or Out of Stock based on configurable reorder levels.
- Generate an availability report listing the current stock status of every product, filterable by status.
- View a dashboard summarizing total products, total stock quantity, and counts of low-stock and out-of-stock items.

4. Plan of Work and Product Ownership

4.1 Plan of Work (Short Term)

Week	Planned Activity
1–2	Finalize the SRS: requirements gathering, functional/non-functional requirement documentation, ER diagram and database schema design.
3	Design UML use-case and class diagrams; set up project repository, technology stack and development environment.
4–5	Implement core inventory module: view/search inventory, add product, delete product, and update current quantity.
6	Implement input validation, duplicate product-ID prevention, and automatic stock-status classification (In Stock / Low Stock / Out of Stock).

Week	Planned Activity
7	Implement availability report generation and filtering.
8	Implement dashboard with summary metrics: total products, total stock quantity, low-stock and out-of-stock counts.
9	Integration testing across modules and UI refinement.
10	Final testing, bug fixes, documentation and demo preparation.

4.2 Product Ownership

The team consists of three members. Functional-feature ownership is assigned according to the major functional features of the proposed system. UI/graphic design, database interaction, unit testing and other common project activities are shared responsibilities across the team and are not counted as functional features.

Team Member	Functional Feature(s) Owned	Qualitative Goal
Shreya C Rao	Login, roles and access control; Add Product; Delete Product; input validation and duplicate product-ID prevention.	Ensure protected operations and secure authentication handling. Ensure data integrity so that no duplicate or invalid product records reach the database.
Shreyas Gowda K A	View/Search Inventory; update current product quantity; automatic stock-status classification (In Stock / Low Stock / Out of Stock).	Ensure efficient inventory search/filter performance and correct availability classification.
R Laksheeth	Availability Report generation and filtering; Dashboard with summary metrics.	Ensure the dashboard and reports accurately reflect current inventory data through a clear, easy-to-read UI.

Scope Clarification

The project maintains the current available quantity for each inventory item. Quantity can be updated when the store's current stock changes, but the system does not record a separate history of Stock In/Stock Out transactions. This keeps Version 1.0 focused on inventory availability, reporting and dashboard functions.